

Vertical-Line Hurwitz Pairings and Exact Stieltjes–Gamma Calculus

Rational dilations, Bernoulli compensation,
polylogarithm identities, and normalized primitives

Research continuation for Vladimir Reshetnikov’s ProveIt project

10 October 2026 • Version 1.0

Abstract

We study ordinary, absolutely convergent bilinear integrals in which the *argument* of a Hurwitz zeta function moves along a vertical line. Subtracting its spatial principal term produces a Laplace kernel whose products close in a finite alphabet of Hurwitz zeta functions. Differentiation at the spectral pole yields an arbitrary-index pairing formula for the compensated Stieltjes functions $\gamma_m(z) + \log^{m+1} z / (m + 1)$. A finite Bernoulli-subtraction calculus extends this closure to negative spectral orders and to normalized antiderivatives of the Stirling remainder. Rationally related vertical scales admit a finite residue-grid reduction; we give an explicit all-order coefficient formula, not merely a meromorphic prescription. Additional identities couple these families to finite generalized harmonic sums, polygamma functions, Lerch transcendents, and polylogarithm order derivatives. Three illustrative specializations evaluate a compensated digamma norm, a Stirling-remainder norm, and a two-scale pairing containing $\log \Gamma(1/4)$. All general identities are proved by Laplace–Plancherel, finite kernel algebra, and holomorphic continuation. Reproducible symbolic and numerical checks are separated from those proofs. A concrete nonreal-argument failure of the installed mpmath 1.3.0 Stieltjes routine is documented and avoided by an independent Cauchy-coefficient calculation. No claim of an exhaustive literature audit, proof-assistant verification, or resolution of the repository’s S_6 and revised S_8 conjectures is made.

Contents

1	Scope, relation to the repository, and principal results	2
2	Laplace–Plancherel and the basic Hurwitz closure	3
3	Arbitrary-index Stieltjes pairings	5
4	Polygamma functions and generalized harmonic sums	8
5	Lerch products and polylogarithm order derivatives	9
6	Bernoulli compensation and negative spectral orders	11
7	Stirling remainders and normalized antiderivatives	13
8	Rationally related vertical scales	15
9	Evaluating removable integer values without numerical cancellation	18
10	Audit findings, corrections, and verification status	18
11	Further research questions	20

1 Scope, relation to the repository, and principal results

The canonical manuscript in [1] already develops a substantial periodic Hurwitz–Stieltjes convolution calculus. In particular, its Fourier-normalized family on $\mathbb{R}/\mathbb{Z}$ leads to shifted correlations, finite parts, coincident limits, contact terms, and primitive ladders. Repeating those formulas under new notation would not advance that program. Here the integration variable is instead the imaginary part of the *second* Hurwitz argument:

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} f(a + iy)g(b - iy) dy, \quad \operatorname{Re} a, \operatorname{Re} b > 0. \quad (1.1)$$

This is neither an integral in the spectral variable $s = \sigma + iy$ nor a periodic correlation on $[0, 1]$. It requires a different subtraction, a different transform, and different convergence hypotheses. The auxiliary Hurwitz products over $[0, 1]$ studied in [5, 6] also have a different integration geometry and use a different auxiliary function. Our starting transform identities are classical; the contribution of this note is the explicit, jointly proved closure calculus and its specializations. We do not assert priority for every displayed identity.

Set

$$Z_s(z) = \zeta(s, z) - \frac{z^{1-s}}{s-1}, \quad G_m(z) = \gamma_m(z) + \frac{\log^{m+1} z}{m+1}. \quad (1.2)$$

The value at $s = 1$ is removable, and $G_0(z) = \log z - \psi(z)$. The main equal-scale theorem evaluates (1.1) for $f = Z_s$, $g = Z_t$ by a one-variable function $K(s + t - 1, a + b)$. Its all-order Taylor coefficients then evaluate every pairing of G_m and G_n . Finite Bernoulli subtractions give the same conclusion for negative spectral orders. Rational vertical dilations introduce only a finite set of shifted Hurwitz arguments, rather than an irreducible double zeta object.

Three consequences indicate the scale and normalization of the results. Define the analytic Stirling remainder in the right half-plane by

$$\Omega(z) = \log \Gamma(z) - (z - \tfrac{1}{2}) \log z + z - \tfrac{1}{2} \log(2\pi). \quad (1.3)$$

Then

$$\frac{1}{2\pi} \int_{\mathbb{R}} |G_0(\tfrac{1}{2} + iy)|^2 dy = \log(2\pi) - \frac{3}{2}, \quad (1.4)$$

$$\frac{1}{2\pi} \int_{\mathbb{R}} |\Omega(\tfrac{1}{2} + iy)|^2 dy = \zeta'(-1) + \frac{\zeta(3)}{2\pi^2} + \frac{1}{9}, \quad (1.5)$$

$$\begin{aligned} \frac{1}{2\pi} \int_{\mathbb{R}} G_0(\tfrac{1}{2} + iy)G_0(\tfrac{1}{2} - 2iy) dy &= \log \Gamma(\tfrac{1}{4}) - \tfrac{1}{2} \log \pi - \tfrac{1}{4} \log 2 \\ &\quad + \tfrac{3}{4} \log 3 - \tfrac{3}{4} - \tfrac{\pi}{8}. \end{aligned} \quad (1.6)$$

Their right-hand sides are approximately 0.33787706640934548, 0.0065868815274467235, and 0.22363092143574916, respectively. These numerical values are diagnostics; the proofs appear below.

1.1 What is and is not resolved

The questions answered here are precise: whether spatially compensated vertical Stieltjes pairings close at arbitrary indices; whether negative spectral orders can be handled without assigning finite parts to the vertical integral; and whether rationally related slopes preserve finite Hurwitz closure. The answer to each is affirmative, with explicit formulas and proofs. These are newly developed continuations in this package, not a claim that a named conjecture in an unread incoming archive has been solved. The repository's S_6 and revised S_8 problems are not settled here.

The audit was targeted. The canonical README, the periodic convolution foundation, a terminal rational-reduction section, the incoming README, and incoming metadata were inspected. Not every recent ZIP was decoded or audited. Thus absence of a formula from these inspected sections is not a proof of absence from the entire corpus. The package includes an integration map and a source inventory so that an independent maintainer can perform that final overlap check.

1.2 Conventions that affect the answer

Throughout, $\log z$ is the principal logarithm on $\operatorname{Re} z > 0$. The function $\log \Gamma(z)$ means its analytic logarithm, real on the positive real axis; it does *not* mean the pointwise principal value of $\log(\Gamma(z))$. A prime on ζ or Φ denotes differentiation in the spectral order, unless a different variable is explicitly indicated. We use

$$\zeta(1+u, z) = \frac{1}{u} + \sum_{m \geq 0} \frac{(-1)^m}{m!} \gamma_m(z) u^m. \quad (1.7)$$

All the vertical integrals asserted as theorems are ordinary absolutely convergent Lebesgue integrals. The words “constant term” and “removable value” used later refer to a finite *spectral expression*, not to a regularization of a divergent integral in y .

2 Laplace–Plancherel and the basic Hurwitz closure

2.1 A bilinear transform lemma

For a function supported on $(0, \infty)$ use the Fourier convention $\widehat{f}(y) = \int_0^\infty f(x) e^{-iyx} dx$. The following version of Plancherel fixes every scale factor needed later.

Lemma 2.1 (Bilinear Laplace–Plancherel). *Suppose $f, g \in L^1(0, \infty) \cap L^2(0, \infty)$. Then*

$$\frac{1}{2\pi} \int_{\mathbb{R}} \widehat{f}(y) \widehat{g}(-y) dy = \int_0^\infty f(x) g(x) dx. \quad (2.1)$$

Both products are absolutely integrable. For $p, q > 0$,

$$\frac{1}{2\pi} \int_{\mathbb{R}} \widehat{f}(py) \widehat{g}(-qy) dy = \int_0^\infty f(qx) g(px) dx. \quad (2.2)$$

If f and g depend holomorphically on parameters as L^2 vectors, all parameter derivatives can be taken inside the bilinear pairing.

Proof. Apply the usual Hermitian Plancherel identity to f and $\overline{g}$, noting that $\overline{\widehat{g}(y)} = \widehat{g}(-y)$. Cauchy–Schwarz gives absolute integrability. For the scaled formula, $\widehat{f}(py)$ is the Fourier transform of $p^{-1}f(x/p)$. Equation (2.1) therefore gives $(pq)^{-1} \int_0^\infty f(x/p) g(x/q) dx$; substitute $x = pqu$. Finally, the bilinear map $(f, g) \mapsto \int fg$ is continuous on $L^2 \times L^2$. Its composition with holomorphic vector-valued maps is holomorphic and satisfies the usual product rule. $\square$

2.2 Spatial compensation

Write

$$Q(x) = \frac{1}{1 - e^{-x}}, \quad h(x) = Q(x) - \frac{1}{x}. \quad (2.3)$$

At the origin, $h(x) = \frac{1}{2} + \frac{x}{12} + O(x^3)$; at infinity it has at most polynomial growth. The standard Mellin representation of the Hurwitz zeta function [2], after subtracting the elementary Laplace transform of x^{s-2} , gives

$$Z_s(z) = \frac{1}{\Gamma(s)} \int_0^\infty x^{s-1} e^{-zx} h(x) dx, \quad \operatorname{Re} s > 0, \quad \operatorname{Re} z > 0. \quad (2.4)$$

Initially one can subtract the two integrals for $\operatorname{Re} s > 1$. The combined integrand is integrable for $\operatorname{Re} s > 0$ and is locally uniformly dominated there, which proves the continuation. In particular, the normalized subtraction is spatial: removing only $1/(s-1)$ would not produce (2.4).

For $\operatorname{Re} s > 1/2$, the function $x^{s-1} e^{-ax} h(x)$ lies in $L^1 \cap L^2$. On each compact subset of this parameter region, every fixed spectral derivative merely adds a power of $\log x$, which is still locally dominated by an

integrable power at zero and an exponential at infinity. This verifies the holomorphic-vector hypothesis of theorem 2.1; it will justify all subsequent Stieltjes jet extractions without a separate unjustified interchange at each order.

Theorem 2.2 (Equal-scale compensated Hurwitz pairing). *Let $\operatorname{Re} a, \operatorname{Re} b > 0$ and $\operatorname{Re} s, \operatorname{Re} t > 1/2$. Put $A = a + b$ and $w = s + t - 1$. Then*

$$\frac{1}{2\pi} \int_{\mathbb{R}} Z_s(a + iy) Z_t(b - iy) dy = \frac{\Gamma(w)}{\Gamma(s)\Gamma(t)} K(w, A), \quad (2.5)$$

where

$$K(w, A) = \left(1 - \frac{2}{w-1}\right) \zeta(w-1, A) + (1-A) \zeta(w, A) + \frac{A^{2-w}}{(w-1)(w-2)}. \quad (2.6)$$

The apparent singularities at $w = 1, 2$ are removable. The combined function $\Gamma(w)K(w, A)$ is holomorphic for $\operatorname{Re} w > 0$ and $\operatorname{Re} A > 0$.

Proof. By (2.4) and theorem 2.1, the integral equals

$$\frac{1}{\Gamma(s)\Gamma(t)} \int_0^\infty x^{w-1} e^{-Ax} h(x)^2 dx. \quad (2.7)$$

For $\operatorname{Re} w > 2$, expand $h^2 = Q^2 - 2Q/x + x^{-2}$. The exact power series

$$Q(x)^2 = \sum_{n \geq 0} (n+1) e^{-nx}$$

yields

$$\int_0^\infty x^{w-1} e^{-Ax} Q(x)^2 dx = \Gamma(w) \{ \zeta(w-1, A) + (1-A) \zeta(w, A) \},$$

because $n+1 = (n+A) + (1-A)$. The other two integrals are $-2\Gamma(w-1)\zeta(w-1, A)$ and $\Gamma(w-2)A^{2-w}$. Using the Gamma recurrence gives (2.6). The combined integral (2.7) is holomorphic for $\operatorname{Re} w > 0$, since $h^2 = 1/4 + O(x)$ at zero. The identity theorem extends the finite expression to that half-plane and removes its apparent poles. $\square$

The pole cancellation can also be checked directly. At $w = 1$ the residue is

$$-2\zeta(0, A) + (1-A) - A = 0, \quad \zeta(0, A) = \frac{1}{2} - A.$$

At $w = 2$, the residue -1 from the first zeta term cancels the residue $+1$ from the elementary term. Neither cancellation is a license to substitute separately into the singular summands.

Corollary 2.3 (An exact compensated digamma integral). *For $\operatorname{Re} a, \operatorname{Re} b > 0$ and $A = a + b$,*

$$\begin{aligned} & \frac{1}{2\pi} \int_{\mathbb{R}} [\log(a + iy) - \psi(a + iy)] [\log(b - iy) - \psi(b - iy)] dy \\ &= \frac{1}{2} - 2A + A \log A - 2 \log \Gamma(A) + \log(2\pi) - (1-A)\psi(A). \end{aligned} \quad (2.8)$$

Proof. The constant term of (2.6) at $w = 1$ is

$$\zeta(0, A) - 2\zeta'(0, A) + (1-A)\gamma_0(A) + A(\log A - 1).$$

Use $\gamma_0 = -\psi$ and $\zeta'(0, A) = \log \Gamma(A) - \frac{1}{2} \log(2\pi)$ [2]. Formula (1.4) follows at $A = 1$. $\square$

2.3 Position derivatives

Direct differentiation gives the exact ladder

$$\partial_z Z_s(z) = -s Z_{s+1}(z), \quad \partial_z^r Z_s(z) = (-1)^r (s)_r Z_{s+r}(z). \quad (2.9)$$

Consequently, for integers $r, k \geq 0$, differentiating (2.5) in a and b gives

$$\begin{aligned} & \frac{1}{2\pi} \int_{\mathbb{R}} (\partial_z^r Z_s)(a + iy) (\partial_z^k Z_t)(b - iy) dy \\ &= \partial_A^{r+k} \frac{\Gamma(w) K(w, A)}{\Gamma(s) \Gamma(t)} = \frac{(-1)^{r+k} \Gamma(w + r + k)}{\Gamma(s) \Gamma(t)} K(w + r + k, A). \end{aligned} \quad (2.10)$$

The last equality follows by differentiating the Laplace moment, and is valid at all its removable spectral values. Thus the differentiation calculus closes without introducing a second independent position parameter. Spectral differentiation of this identity gives the same result for derivatives of arbitrary G_m .

3 Arbitrary-index Stieltjes pairings

3.1 The complete coefficient theorem

Expansion of the subtracted term in (1.2) gives

$$Z_{1+u}(z) = \sum_{m \geq 0} \frac{(-1)^m}{m!} G_m(z) u^m. \quad (3.1)$$

For $\operatorname{Re} a, \operatorname{Re} b > 0$, define

$$I_{m,n}(A) = \frac{1}{2\pi} \int_{\mathbb{R}} G_m(a + iy) G_n(b - iy) dy, \quad A = a + b. \quad (3.2)$$

The following theorem both proves that this depends only on A and specifies a finite answer at every pair of indices.

Theorem 3.1 (All-index Stieltjes closure). *For all integers $m, n \geq 0$, the integral (3.2) converges absolutely and*

$$I_{m,n}(A) = (-1)^{m+n} m! n! [u^m v^n] C(u, v) K(1 + u + v, A), \quad (3.3)$$

where

$$C(u, v) = \frac{\Gamma(1 + u + v)}{\Gamma(1 + u) \Gamma(1 + v)}, \quad (3.4)$$

$$\log C(u, v) = \sum_{r \geq 2} \frac{(-1)^r \zeta(r)}{r} \{(u + v)^r - u^r - v^r\}. \quad (3.5)$$

Write $K(1 + \epsilon, A) = \sum_{\ell \geq 0} k_\ell(A) \epsilon^\ell$. Its coefficients are

$$\begin{aligned} k_\ell(A) &= \frac{\zeta^{(\ell)}(0, A)}{\ell!} - \frac{2\zeta^{(\ell+1)}(0, A)}{(\ell + 1)!} + (1 - A) \frac{(-1)^\ell \gamma_\ell(A)}{\ell!} \\ &\quad - A \sum_{j=0}^{\ell+1} \frac{(-\log A)^j}{j!}. \end{aligned} \quad (3.6)$$

In particular, $I_{m,n}$ belongs to the finite algebra generated by A , $\log A$, the constants $\zeta(2), \dots, \zeta(m + n)$, $\gamma_\ell(A)$ for $\ell \leq m + n$, and $\zeta^{(j)}(0, A)$ for $j \leq m + n + 1$.

Proof. The map $u \mapsto Z_{1+u}(a + iy)$ is holomorphic as an $L^2(\mathbb{R})$ vector in a neighborhood of zero, by the estimates preceding theorem 2.2. Its derivatives are precisely $(-1)^m G_m$. Hence every G_m belongs to L^2 , and the product in (3.2) is absolutely integrable. Taking the two Taylor coefficients in (2.5) gives (3.3). The Taylor expansion of $\log \Gamma(1 + u)$ proves (3.5); its linear Euler-constant terms cancel.

For the explicit coefficients, write

$$K(1 + \epsilon, A) = \left(1 - \frac{2}{\epsilon}\right) \zeta(\epsilon, A) + (1 - A) \zeta(1 + \epsilon, A) + \frac{A e^{-\epsilon \log A}}{\epsilon(\epsilon - 1)}.$$

The pole coefficient vanishes as already checked. Since

$$\frac{A e^{-\epsilon \log A}}{\epsilon(\epsilon - 1)} = -A \sum_{r \geq 0} \epsilon^{r-1} \sum_{j=0}^r \frac{(-\log A)^j}{j!},$$

its coefficient of ϵ^ℓ is the last term of (3.6). The other terms follow from the Taylor and Laurent series of the two zeta factors. Only total degree at most $m + n$ of C is used, proving the finite-alphabet assertion. $\square$

This theorem is an actual finite recipe. If $C(u, v) = \sum c_{r,k} u^r v^k$, then equivalently

$$I_{m,n}(A) = (-1)^{m+n} m! n! \sum_{r=0}^m \sum_{k=0}^n c_{r,k} \binom{m+n-r-k}{m-r} k_{m+n-r-k}(A). \quad (3.7)$$

No two-variable special function or undetermined integration constant remains. This is a closure statement, not a claim that all its generators are algebraically independent or reducible to elementary constants.

3.2 Low-order identities and a useful degeneration

Let $\mu_j(A) = \partial_w^j K(1, A) = j! k_j(A)$. Direct coefficient extraction from (3.5) gives

$$I_{0,0} = \mu_0, \quad I_{m,0} = (-1)^m \mu_m, \quad (3.8)$$

$$I_{1,1} = \mu_2 + \zeta(2) \mu_0, \quad (3.9)$$

$$I_{2,1} = -\mu_3 - 2\zeta(2) \mu_1 + 2\zeta(3) \mu_0, \quad (3.10)$$

$$I_{3,1} = \mu_4 + 3\zeta(2) \mu_2 - 6\zeta(3) \mu_1 + 6\zeta(4) \mu_0, \quad (3.11)$$

$$I_{2,2} = \mu_4 + 4\zeta(2) \mu_2 - 8\zeta(3) \mu_1 + \{6\zeta(4) + 2\zeta(2)^2\} \mu_0. \quad (3.12)$$

Here and below, every μ_j in one equation has the same argument A . For example, the coefficient of $u^2 v^2$ in C is $\frac{3}{2}\zeta(4) + \frac{1}{2}\zeta(2)^2$; the factor $2!2!$ explains the last coefficient in (3.12). This factorial is easy to lose if ordinary and exponential generating functions are mixed.

At $A = 1$, the entire explicit Stieltjes term in (3.6) disappears:

$$k_\ell(1) = \frac{\zeta^{(\ell)}(0)}{\ell!} - \frac{2\zeta^{(\ell+1)}(0)}{(\ell+1)!} - 1. \quad (3.13)$$

Thus every vertical pairing with $a + b = 1$ is expressible using Riemann zeta order derivatives at zero and ordinary positive zeta values. This does not say that the individual Stieltjes functions disappear pointwise; the simplification occurs after pairing.

3.3 A second, kernel-polynomial description

Define real-coefficient polynomials $E_m(L)$ by

$$\sum_{m \geq 0} E_m(L) \frac{v^m}{m!} = \frac{e^{-vL}}{\Gamma(1-v)}. \quad (3.14)$$

For example,

$$E_0 = 1, \quad E_1 = -(L + \gamma), \quad E_2 = (L + \gamma)^2 - \zeta(2).$$

Differentiating (2.4) gives

$$G_m(z) = \int_0^\infty e^{-zx} h(x) E_m(\log x) dx, \quad (3.15)$$

$$I_{m,n}(A) = \int_0^\infty e^{-Ax} h(x)^2 E_m(\log x) E_n(\log x) dx. \quad (3.16)$$

All these differentiations are justified by the same compact-parameter bounds used in the proof of theorem 3.1. Equations (3.3) and (3.16) provide two useful computational routes: the former uses a finite zeta/Stieltjes alphabet, while the latter uses one rapidly decaying quadrature with a kernel regular for positive x . They are independently implemented in the verification code.

The equal-scale identity also supplies all derivatives in the horizontal arguments: one differentiates (3.3) in A , or inserts powers of $-x$ in (3.16). No new spectral-order restriction arises from a nonnegative number of such derivatives.

3.4 A complex-safe Hermite identity

The generalized Stieltjes parameter must be treated holomorphically, not by applying a real-part formula outside its real domain. A useful exact representation, valid at every index, is the following.

Proposition 3.2 (Holomorphic Hermite representation). *For $\operatorname{Re} z > 0$ and every integer $m \geq 0$,*

$$G_m(z) = \frac{\log^m z}{2z} + i \int_0^\infty \frac{1}{e^{2\pi t} - 1} \left\{ \frac{\log^m(z + it)}{z + it} - \frac{\log^m(z - it)}{z - it} \right\} dt. \quad (3.17)$$

The integral is absolutely convergent and holomorphic in z throughout the right half-plane. For nonreal z , neither term may be replaced by the complex conjugate of the other.

Proof. The standard Hermite representation of Hurwitz zeta [2], after subtracting its spatial principal term, is

$$Z_s(z) = \frac{1}{2z^s} + i \int_0^\infty \frac{(z + it)^{-s} - (z - it)^{-s}}{e^{2\pi t} - 1} dt.$$

It can equivalently be obtained from (2.4) using $h(x) = 1/2 + 2 \int_0^\infty \sin(tx)/(e^{2\pi t} - 1) dt$. The two powers have a difference $O(t)$ at zero, uniformly on compact subsets of $s \in \mathbb{C}$, $\operatorname{Re} z > 0$; the denominator also has a simple zero. At infinity the exponential denominator dominates all powers and logarithms arising from any fixed spectral derivative. Therefore the combined integral is jointly holomorphic and may be differentiated in s at $s = 1$. Multiplying the m th derivative by $(-1)^m$ proves (3.17). Holomorphic continuation in z from positive real values gives the entire right half-plane; the paired expression, not a real-part projection, is what makes this continuation legitimate. $\square$

This representation is classical in origin, but spelling out its paired complex form is important for the verification of the present vertical-line identities. Section 10.2 records a concrete numerical failure when a real-part implementation is used at nonreal z , and supplies both (3.17) and Cauchy-coefficient extraction as repairs.

4 Polygamma functions and generalized harmonic sums

4.1 Uncompensated polygamma pairings

For uncompensated Hurwitz zeta functions, the Laplace kernel $Q(x)$ has a simple pole at zero. Consequently, $\operatorname{Re} s, \operatorname{Re} t > 3/2$ is a sufficient L^2 region for the same argument to give

$$\frac{1}{2\pi} \int_{\mathbb{R}} \zeta(s, a + iy) \zeta(t, b - iy) dy = \frac{\Gamma(w)}{\Gamma(s)\Gamma(t)} \{\zeta(w-1, A) + (1-A)\zeta(w, A)\}. \quad (4.1)$$

This follows directly from the Q^2 calculation in the proof of theorem 2.2; no compensation is needed in this smaller domain. Using the standard identity $\psi^{(r)}(z) = (-1)^{r+1} r! \zeta(r+1, z)$ gives the following consequence.

Corollary 4.1. *For integers $r, k \geq 1$, $\operatorname{Re} a, \operatorname{Re} b > 0$, and $A = a + b$,*

$$\begin{aligned} \frac{1}{2\pi} \int_{\mathbb{R}} \psi^{(r)}(a + iy) \psi^{(k)}(b - iy) dy \\ = (-1)^{r+k} (r+k)! \{\zeta(r+k, A) + (1-A)\zeta(r+k+1, A)\}. \end{aligned} \quad (4.2)$$

In particular,

$$\frac{1}{2\pi} \int_{\mathbb{R}} |\psi^{(r)}(\tfrac{1}{2} + iy)|^2 dy = (2r)! \zeta(2r). \quad (4.3)$$

The restriction $r, k \geq 1$ matters. Setting both to zero would introduce the divergent squared norm of the uncorrected digamma function, not the finite value in (1.4).

4.2 Finite generalized harmonic families

For a positive integer N , define

$$\mathcal{H}_N^{(s)}(z) = \sum_{j=0}^{N-1} (z+j)^{-s}. \quad (4.4)$$

At $z = 1$ this is the usual generalized harmonic number $H_N^{(s)}$. For $\operatorname{Re} s > 0$, it is the Laplace transform with kernel $\sum_{j=0}^{N-1} e^{-jx}$, divided by $\Gamma(s)$.

Theorem 4.2 (Finite harmonic and mixed pairings). *Let $M, N \geq 1$, $\operatorname{Re} a, \operatorname{Re} b > 0$, $\operatorname{Re} s, \operatorname{Re} t > 1/2$, and $A = a + b$, $w = s + t - 1$. Then*

$$\begin{aligned} \frac{1}{2\pi} \int_{\mathbb{R}} \mathcal{H}_M^{(s)}(a + iy) \mathcal{H}_N^{(t)}(b - iy) dy \\ = \frac{\Gamma(w)}{\Gamma(s)\Gamma(t)} \sum_{j=0}^{M-1} \sum_{k=0}^{N-1} (A + j + k)^{-w}, \end{aligned} \quad (4.5)$$

$$\begin{aligned} \frac{1}{2\pi} \int_{\mathbb{R}} Z_s(a + iy) \mathcal{H}_N^{(t)}(b - iy) dy \\ = \frac{\Gamma(w)}{\Gamma(s)\Gamma(t)} \sum_{k=0}^{N-1} Z_w(A + k). \end{aligned} \quad (4.6)$$

The second expression is taken at its removable value if $w = 1$. Arbitrary spectral and position derivatives are permitted inside both identities in the stated domain.

Proof. Multiply the two finite exponential kernels and apply theorem 2.1. Each exponential term contributes $\Gamma(w)(A + j + k)^{-w}$. In the mixed product it contributes $\Gamma(w)Z_w(A + k)$ by (2.4). The kernels are bounded at zero, so the same L^2 holomorphy argument applies to all derivatives. $\square$

For $a + b = 1$ and $s = t = 1$, these become particularly elementary:

$$\begin{aligned} \frac{1}{2\pi} \int_{\mathbb{R}} \mathcal{H}_M^{(1)}(a + iy) \mathcal{H}_N^{(1)}(b - iy) dy \\ = (M + N)H_{M+N} - MH_M - NH_N, \end{aligned} \quad (4.7)$$

$$\begin{aligned} \frac{1}{2\pi} \int_{\mathbb{R}} G_0(a + iy) \mathcal{H}_N^{(1)}(b - iy) dy \\ = \log \Gamma(N + 1) - NH_N + N(1 + \gamma). \end{aligned} \quad (4.8)$$

Indeed, the first double sum can be summed along its diagonals, and the second is $\sum_{k=1}^N (\log k - \psi(k))$. Taking spectral derivatives supplies finite sums of $\log^r(A + j + k)/(A + j + k)^w$ and corresponding exact mixed identities for higher G_m , with the same Gamma-quotient coefficient bookkeeping as in section 3.

5 Lerch products and polylogarithm order derivatives

Let ξ lie in $\{\xi : |\xi| \leq 1, \xi \neq 1\}$. The Lerch transcendent has the integral representation [4]

$$\Phi(\xi, s, z) = \frac{1}{\Gamma(s)} \int_0^\infty \frac{x^{s-1} e^{-zx}}{1 - \xi e^{-x}} dx, \quad \operatorname{Re} s > 0, \quad \operatorname{Re} z > 0. \quad (5.1)$$

For $|\xi| < 1$ this also follows by summing its defining series. The kernel is bounded at zero and along the positive real axis. Boundary colors $|\xi| = 1, \xi \neq 1$, follow by dominated continuation from inside the disk. No assertion here includes the singular color $\xi = 1$ without the separate compensation already developed.

Theorem 5.1 (Colored pairings and confluent colors). *Let $\operatorname{Re} s, \operatorname{Re} t > 1/2$, $\operatorname{Re} a, \operatorname{Re} b > 0$, and set $A = a + b$, $w = s + t - 1$. For admissible colors $\xi \neq \eta$,*

$$\begin{aligned} \frac{1}{2\pi} \int_{\mathbb{R}} \Phi(\xi, s, a + iy) \Phi(\eta, t, b - iy) dy \\ = \frac{\Gamma(w)}{\Gamma(s)\Gamma(t)} \frac{\xi \Phi(\xi, w, A) - \eta \Phi(\eta, w, A)}{\xi - \eta}. \end{aligned} \quad (5.2)$$

At $\eta = \xi$, the quotient is replaced by

$$\Phi(\xi, w - 1, A) + (1 - A)\Phi(\xi, w, A). \quad (5.3)$$

In particular, when $A = 1$ and $\xi \neq \eta$,

$$\begin{aligned} \frac{1}{2\pi} \int_{\mathbb{R}} \Phi(\xi, s, a + iy) \Phi(\eta, t, 1 - a - iy) dy \\ = \frac{\Gamma(w)}{\Gamma(s)\Gamma(t)} \frac{\operatorname{Li}_w(\xi) - \operatorname{Li}_w(\eta)}{\xi - \eta}. \end{aligned} \quad (5.4)$$

For equal nonzero colors the final quotient is $\operatorname{Li}_{w-1}(\xi)/\xi$; at $\xi = 0$ its removable value is 1.

Proof. For distinct colors, use the exact partial fraction identity

$$\frac{1}{(1 - \xi r)(1 - \eta r)} = \frac{\xi/(1 - \xi r) - \eta/(1 - \eta r)}{\xi - \eta}$$

in the Laplace product supplied by theorem 2.1. For equal colors, the kernel has series $\sum_{n \geq 0} (n+1)\xi^n e^{-nx}$, and $n+1 = (n+A) + (1-A)$ proves (5.3). This series argument initially uses $|\xi| < 1$; its identity extends to the stated boundary colors by the same dominated kernel argument. Finally, $\xi \Phi(\xi, w, 1) = \operatorname{Li}_w(\xi)$, including the removable interpretation at zero. All products converge by L^2 . $\square$

Order differentiation of (5.4) is a direct source of identities for $\partial_w^r \text{Li}_w(\xi)$ and mixed order derivatives of Lerch functions. In particular, at $s = t = 1$, the total-order derivatives are generated by

$$C(u, v) \frac{\text{Li}_{1+u+v}(\xi) - \text{Li}_{1+u+v}(\eta)}{\xi - \eta}.$$

This is a convergent parameter identity near the origin, rather than an experimental relation inferred from numerical integer-relation searches.

5.1 Mixed Stieltjes–Lerch closure and the spectral resonance

Theorem 5.2. *Under the hypotheses of theorem 5.1,*

$$\begin{aligned} & \frac{1}{2\pi} \int_{\mathbb{R}} Z_s(a + iy) \Phi(\xi, t, b - iy) dy \\ &= \frac{\Gamma(w)}{\Gamma(s)\Gamma(t)} \left\{ \frac{\zeta(w, A) - \xi \Phi(\xi, w, A)}{1 - \xi} - \frac{\Phi(\xi, w - 1, A)}{w - 1} \right\}. \end{aligned} \quad (5.5)$$

The brace is holomorphic for $\text{Re } w > 0$. At $s = t = 1$ it gives

$$\begin{aligned} & \frac{1}{2\pi} \int_{\mathbb{R}} G_0(a + iy) \Phi(\xi, 1, b - iy) dy \\ &= \frac{-\psi(A) - \xi \Phi(\xi, 1, A)}{1 - \xi} - \Phi'(\xi, 0, A). \end{aligned} \quad (5.6)$$

Proof. The product kernel is

$$\frac{h(x)}{1 - \xi e^{-x}} = \frac{Q(x) - \xi/(1 - \xi e^{-x})}{1 - \xi} - \frac{1}{x(1 - \xi e^{-x})}.$$

Integrate the separated expression first for $\text{Re } w > 1$. The combined kernel is bounded at zero, so its Mellin transform is holomorphic for $\text{Re } w > 0$. At $w = 1$, both singular summands have residue $1/(1 - \xi)$ with opposite signs, because $\Phi(\xi, 0, A) = 1/(1 - \xi)$. Their constant terms give (5.6). Derivatives of this removable identity are justified by the original holomorphic L^2 pairing. $\square$

When $A = 1$, (5.6) becomes

$$\frac{1}{2\pi} \int_{\mathbb{R}} G_0(a + iy) \Phi(\xi, 1, 1 - a - iy) dy = \frac{\gamma + \log(1 - \xi)}{1 - \xi} - \frac{\partial_s \text{Li}_s(\xi)|_{s=0}}{\xi}. \quad (5.7)$$

At zero color the final quotient is interpreted by continuity. At $\xi = -1$, the right-hand side reduces to

$$\frac{\gamma}{2} + \log 2 - \frac{1}{2} \log \pi. \quad (5.8)$$

Indeed, $\text{Li}_s(-1) = (2^{1-s} - 1)\zeta(s)$, whence $\text{Li}'_0(-1) = \frac{1}{2} \log(2/\pi)$. Moreover,

$$\Phi(-1, 1, z) = \frac{1}{2} \left\{ \psi\left(\frac{z+1}{2}\right) - \psi\left(\frac{z}{2}\right) \right\}.$$

Thus (5.8) is also an explicit compensated digamma–digamma-difference integral.

5.2 Every nontrivial root of unity gives a finite Gamma formula

If $\xi^q = 1$ and $\xi \neq 1$, grouping the Lerch series modulo q gives

$$\Phi(\xi, s, A) = q^{-s} \sum_{r=0}^{q-1} \xi^r \zeta\left(s, \frac{A+r}{q}\right). \quad (5.9)$$

The zeta residues at $s = 1$ cancel because $\sum \xi^r = 0$. With $c_r = (A+r)/q$, its two required values are

$$\Phi(\xi, 1, A) = -\frac{1}{q} \sum_{r=0}^{q-1} \xi^r \psi(c_r), \quad (5.10)$$

$$\Phi'(\xi, 0, A) = \sum_{r=0}^{q-1} \xi^r \log \Gamma(c_r) - \frac{\log q}{1-\xi}. \quad (5.11)$$

Substituting into (5.6) yields the finite family

$$\begin{aligned} & \frac{1}{2\pi} \int_{\mathbb{R}} G_0(a+iy) \Phi(\xi, 1, b-iy) dy \\ &= \frac{-\psi(A) + (\xi/q) \sum_{r=0}^{q-1} \xi^r \psi(c_r) + \log q}{1-\xi} - \sum_{r=0}^{q-1} \xi^r \log \Gamma(c_r). \end{aligned} \quad (5.12)$$

For higher indices, differentiating the finite sum (5.9) gives a triangular combination of Hurwitz order jets and powers of $\log q$. This is a proved finite reduction at every index, with no presumption of further transcendence reductions.

6 Bernoulli compensation and negative spectral orders

6.1 A hierarchy with an exact convergence domain

The basic kernel h is bounded but does not vanish at zero. Its Mellin transform therefore does not directly reach the Gamma logarithm at spectral order zero. We now subtract exactly the required Taylor terms. Let B_{2k} be the Bernoulli numbers and put

$$P_0(x) = x^{-1}, \quad (6.1)$$

$$P_d(x) = x^{-1} + \frac{1}{2} + \sum_{k=1}^{d-1} \frac{B_{2k}}{(2k)!} x^{2k-1}, \quad d \geq 1, \quad (6.2)$$

$$h_d(x) = Q(x) - P_d(x), \quad \nu_d = \begin{cases} 0, & d = 0, \\ 2d-1, & d \geq 1. \end{cases} \quad (6.3)$$

An empty sum is zero. Thus $h_0 = h$, $h_1 = h - 1/2$, and $h_2 = h - 1/2 - x/12$. The convergent local Bernoulli expansion gives

$$h_0(x) = \frac{1}{2} + O(x), \quad h_d(x) = \frac{B_{2d}}{(2d)!} x^{2d-1} + O(x^{2d+1}) \quad (d \geq 1). \quad (6.4)$$

Write $P_d(x) = \sum_{j \in E_d} \alpha_{d,j} x^j$, where $E_0 = \{-1\}$ and $E_d = \{-1, 0, 1, 3, \dots, 2d-3\}$ for $d \geq 1$, with the evident empty positive portion for $d = 1$.

Definition 6.1. The Gamma-normalized compensated Hurwitz family is

$$\mathcal{F}_d(s, z) = \int_0^\infty x^{s-1} e^{-zx} h_d(x) dx, \quad \operatorname{Re} s > -\nu_d, \quad \operatorname{Re} z > 0. \quad (6.5)$$

Equivalently, on this domain it is the combined continuation of

$$\mathcal{F}_d(s, z) = \Gamma(s)\zeta(s, z) - \sum_{j \in E_d} \alpha_{d,j} \Gamma(s+j) z^{-s-j}. \quad (6.6)$$

We also set $\mathcal{R}_d(s, z) = \mathcal{F}_d(s, z)/\Gamma(s)$ by continuation. In particular,

$$\begin{aligned} \mathcal{R}_0(s, z) &= Z_s(z), \\ \mathcal{R}_d(s, z) &= Z_s(z) - \frac{1}{2}z^{-s} - \sum_{k=1}^{d-1} \frac{B_{2k}}{(2k)!} (s)_{2k-1} z^{1-s-2k} \quad (d \geq 1). \end{aligned} \quad (6.7)$$

Proposition 6.2 (Holomorphy, zeros, and the position ladder). *The family $\mathcal{F}_d$ is jointly holomorphic on the domain in (6.5). It is a holomorphic L^2 vector on the vertical line $z = a + iy$ whenever*

$$\operatorname{Re} s > \frac{1}{2} - \nu_d, \quad \operatorname{Re} a > 0. \quad (6.8)$$

It satisfies

$$\partial_z \mathcal{F}_d(s, z) = -\mathcal{F}_d(s+1, z). \quad (6.9)$$

For every integer n with $0 \leq n < \nu_d$,

$$\mathcal{R}_d(-n, z) = 0, \quad \mathcal{F}_d(-n, z) = \frac{(-1)^n}{n!} \partial_s \mathcal{R}_d(s, z)|_{s=-n}. \quad (6.10)$$

Proof. Near zero the kernel in (6.5) is $O(x^{\operatorname{Re} s + \nu_d - 1})$, and at infinity the exponential dominates all polynomial terms in P_d . These bounds, with extra powers of $\log x$ on compact spectral subsets, prove holomorphy. Squaring the near-zero kernel gives exactly the sufficient L^2 condition (6.8). Plancherel then transfers the vector-valued holomorphy to the vertical line. Multiplication by $-x$ proves the position ladder.

Formula (6.6) follows by evaluating the separate terms first for sufficiently large $\operatorname{Re} s$, then continuing. The function in (6.7) is entire in s , after the original $s = 1$ pole cancellation. Since $1/\Gamma(s)$ has a simple zero at $s = -n$ while $\mathcal{F}_d$ is holomorphic there, $\mathcal{R}_d(-n, z) = 0$. Finally, $\Gamma(-n + \epsilon) = (-1)^n/(n!\epsilon) + O(1)$ gives the second formula. $\square$

The kernel L^2 threshold is exact for an individual transform in its Laplace domain, because the leading coefficients in (6.4) are nonzero. We do not claim that the rectangular pair of such thresholds is the maximal possible region for every product. The theorem deliberately states a uniform domain in which all orders of parameter differentiation are justified.

6.2 Finite closure at every compensation depth

Define

$$\mathcal{J}_{d,e}(w, A) = \int_0^\infty x^{w-1} e^{-Ax} h_d(x) h_e(x) dx. \quad (6.11)$$

It is holomorphic for $\operatorname{Re} w > -\nu_d - \nu_e$, $\operatorname{Re} A > 0$.

Theorem 6.3 (Arbitrary-depth product closure). *For $\operatorname{Re} s > 1/2 - \nu_d$, $\operatorname{Re} t > 1/2 - \nu_e$, and $\operatorname{Re} a, \operatorname{Re} b > 0$,*

$$\frac{1}{2\pi} \int_{\mathbb{R}} \mathcal{F}_d(s, a + iy) \mathcal{F}_e(t, b - iy) dy = \mathcal{J}_{d,e}(s+t-1, a+b). \quad (6.12)$$

The right-hand side has the finite expression

$$\begin{aligned} \mathcal{J}_{d,e}(w, A) = & \Gamma(w) \{ \zeta(w-1, A) + (1-A)\zeta(w, A) \} \\ & - \sum_{j \in E_d} \alpha_{d,j} \Gamma(w+j) \zeta(w+j, A) \\ & - \sum_{k \in E_e} \alpha_{e,k} \Gamma(w+k) \zeta(w+k, A) \\ & + \sum_{j \in E_d} \sum_{k \in E_e} \alpha_{d,j} \alpha_{e,k} \Gamma(w+j+k) A^{-w-j-k}. \end{aligned} \quad (6.13)$$

Every apparent pole of this expression in $\operatorname{Re} w > -v_d - v_e$ is removable. The identity and all its spectral and position derivatives are identities of ordinary convergent pairings.

Proof. Equation (6.12) is theorem 2.1 applied to (6.5). For $\operatorname{Re} w > 2$, expand

$$(Q - P_d)(Q - P_e) = Q^2 - P_d Q - P_e Q + P_d P_e.$$

Every individual Mellin integral then converges. The Q^2 term was evaluated earlier; the monomial- Q terms yield $\Gamma(w+j)\zeta(w+j, A)$, and the monomial products yield the last line of (6.13). The original combined integrand vanishes to order $v_d + v_e$ at zero, proving its holomorphy on the larger half-plane. Analytic continuation therefore cancels *all* of the apparent poles there, not merely those checked at finitely many indices. The vector-valued argument proves the derivative assertion. $\square$

For $d = e = 0$, (6.13) is exactly $\mathcal{J}_{0,0}(w, A) = \Gamma(w)K(w, A)$. Higher depths do not replace the special-function alphabet: they enlarge its finite range of spectral arguments while improving the admissible left half-plane. This is the mechanism that permits antiderivatives to be paired without a divergent vertical integral.

7 Stirling remainders and normalized antiderivatives

7.1 The first negative-order pairing

At depth one, (6.10) and the standard zeta values at zero give

$$\mathcal{F}_1(0, z) = \Omega(z) = \int_0^\infty e^{-zx} \frac{h_1(x)}{x} dx. \quad (7.1)$$

For completeness, differentiate $\mathcal{R}_1(s, z) = \zeta(s, z) - z^{1-s}/(s-1) - \frac{1}{2}z^{-s}$ at $s = 0$. The result is $\zeta'(0, z) - z \log z + z + \frac{1}{2} \log z$, which is (1.3). Thus (7.1) also follows directly from the compensated Hurwitz family, independently of quoting Binet's formula; it agrees with the classical representation in [3].

Theorem 7.1 (Exact Stirling-remainder pairing). *Let $\operatorname{Re} a, \operatorname{Re} b > 0$ and $A = a + b$. Then*

$$\begin{aligned} \frac{1}{2\pi} \int_{\mathbb{R}} \Omega(a + iy) \Omega(b - iy) dy = & A \zeta'(-1, A) - 2 \zeta'(-2, A) \\ & + \left(\frac{A^3}{6} - \frac{A^2}{2} + \frac{A}{4} \right) \log A + \frac{A^3}{36} + \frac{A}{12}. \end{aligned} \quad (7.2)$$

Proof. The domain condition (6.8) holds at $d = 1, s = 0$. Hence the left-hand side equals $\mathcal{J}_{1,1}(-1, A)$. Since $h_1^2 = h^2 - h + 1/4$, write $\mathcal{J}_{1,1}(w, A) = \Gamma(w) \mathcal{B}(w, A)$, where

$$\begin{aligned} \mathcal{B}(w, A) = & \frac{w-3}{w-1} \zeta(w-1, A) - A \zeta(w, A) + \frac{A^{2-w}}{(w-1)(w-2)} \\ & + \frac{A^{1-w}}{w-1} + \frac{1}{4} A^{-w}. \end{aligned} \quad (7.3)$$

At $w = -1$,

$$\mathcal{B}(-1, A) = 2\zeta(-2, A) - A\zeta(-1, A) + \frac{A^3}{6} - \frac{A^2}{2} + \frac{A}{4} = 0.$$

This is the Bernoulli-polynomial identity $\zeta(-n, A) = -B_{n+1}(A)/(n+1)$ at $n = 1, 2$. Because $\text{Res}_{w=-1} \Gamma(w) = -1$,

$$\mathcal{J}_{1,1}(-1, A) = -\partial_w \mathcal{B}(-1, A).$$

Differentiation before simplification gives

$$\begin{aligned} -\partial_w \mathcal{B}(-1, A) &= A\zeta'(-1, A) - 2\zeta'(-2, A) - \frac{1}{2}\zeta(-2, A) \\ &\quad + A^3 \left(\frac{\log A}{6} - \frac{5}{36} \right) - \frac{A^2 \log A}{2} + \frac{A^2}{4} + \frac{A \log A}{4}. \end{aligned}$$

Finally, $\zeta(-2, A) = -A^3/3 + A^2/2 - A/6$ yields (7.2). $\square$

At $A = 1$, the functional equation of the Riemann zeta function gives $\zeta'(-2) = -\zeta(3)/(4\pi^2)$, proving (1.5). The sign in front of $\partial_w \mathcal{B}$ is forced by the negative residue of Γ at -1 . Discarding the zero of $\mathcal{B}$ before differentiating would incorrectly turn a nonzero norm into zero.

7.2 A canonical primitive at every admissible depth

Define the depth- d Stirling remainder, for $d \geq 1$, by

$$\Omega_d(z) = \Omega(z) - \sum_{k=1}^{d-1} \frac{B_{2k}}{2k(2k-1)} z^{1-2k}. \quad (7.4)$$

The Gamma recurrence in (6.6) gives $\mathcal{F}_d(0, z) = \Omega_d(z)$.

Theorem 7.2 (Normalized primitive ladder and exact pairings). *For $d \geq 1$ and $0 \leq n < \nu_d = 2d - 1$, let*

$$\mathcal{P}_{d,n}(z) = (-1)^n \mathcal{F}_d(-n, z) = \frac{1}{n!} \partial_s \mathcal{R}_d(s, z)|_{s=-n}. \quad (7.5)$$

Then

$$\mathcal{P}_{d,0} = \Omega_d, \quad \partial_z \mathcal{P}_{d,n} = \mathcal{P}_{d,n-1} \quad (n \geq 1). \quad (7.6)$$

Each $\mathcal{P}_{d,n}(x)$ tends to zero as real $x \rightarrow +\infty$. This normalization uniquely specifies it among the holomorphic n -fold antiderivatives of Ω_d . For $d, e \geq 1$, $0 \leq n < \nu_d$, $0 \leq m < \nu_e$,

$$\frac{1}{2\pi} \int_{\mathbb{R}} \mathcal{P}_{d,n}(a + iy) \mathcal{P}_{e,m}(b - iy) dy = (-1)^{n+m} \mathcal{J}_{d,e}(-n - m - 1, a + b). \quad (7.7)$$

The right-hand side is the removable value of the finite expression (6.13); no new integral is left unevaluated.

Proof. The definition and ladder follow from theorem 6.2. The Laplace representation is

$$\mathcal{P}_{d,n}(z) = (-1)^n \int_0^\infty x^{-n-1} e^{-zx} h_d(x) dx.$$

Its near-zero power is x^{ν_d-n-1} , with $\nu_d - n > 0$. Rescaling x by a positive real z and using an integrable split of the kernel proves decay $O(z^{n-\nu_d})$ as $z \rightarrow +\infty$. The difference of two n -fold antiderivatives is a polynomial of degree less than n , and such a polynomial cannot tend to zero unless it is zero. For $n = 0$ the uniqueness statement simply identifies the function itself.

The integer restriction $n < \nu_d$ implies $-n > 1/2 - \nu_d$; similarly for m . Thus theorem 6.3 applies and proves (7.7). Moreover, $-n - m - 1 > -\nu_d - \nu_e$, placing the spectral value strictly inside the holomorphic domain of $\mathcal{J}_{d,e}$. $\square$

The primitive formula is explicit in ordinary Hurwitz order derivatives: (6.7) involves only $\zeta(s, z)$, elementary powers, and a finite Pochhammer polynomial. At $s = -n$ its first derivative therefore involves $\zeta'(-n, z)$ and elementary terms. Pairing primitives at arbitrary depths remains finite by (6.13). At its integer resonances, first derivatives of Hurwitz zeta and digamma values suffice; the next section gives a stable finite-part rule. Higher *spectral* jets may require correspondingly higher zeta derivatives and Stieltjes constants, but always only finitely many.

8 Rationally related vertical scales

Unequal slopes introduce two arithmetic progressions into the product kernel. For a rational slope ratio those progressions are commensurate, and their apparent double-zeta structure reduces to a finite collection of single Hurwitz zeta functions. The scale factors in this statement are important: one cannot obtain it by replacing $a + b$ with a weighted sum in theorem 2.2 while leaving all other factors unchanged.

Let p, q be positive integers, not necessarily coprime. Put

$$B = qa + pb, \quad R = pq, \quad c_{ij} = \frac{B + qi + pj}{R} \quad (0 \leq i < p, 0 \leq j < q), \quad (8.1)$$

and define

$$D_{p,q}(w, B) = R^{-w} \sum_{i=0}^{p-1} \sum_{j=0}^{q-1} \{\zeta(w-1, c_{ij}) + (1 - c_{ij})\zeta(w, c_{ij})\}. \quad (8.2)$$

Lemma 8.1 (Finite residue-grid reduction). *For $\operatorname{Re} w > 2$ and $\operatorname{Re} B > 0$,*

$$\int_0^\infty x^{w-1} e^{-Bx} Q(qx)Q(px) dx = \Gamma(w) D_{p,q}(w, B). \quad (8.3)$$

Proof. Write each pair of nonnegative integers uniquely as $n = p\ell + i$, $m = qk + j$ with $0 \leq i < p$, $0 \leq j < q$. Then $qn + pm = qi + pj + pq(\ell + k)$, so

$$Q(qx)Q(px) = \sum_{i=0}^{p-1} \sum_{j=0}^{q-1} \frac{e^{-(qi+pj)x}}{(1 - e^{-Rx})^2}.$$

Expanding the square produces the multiplicity $\ell + k + 1$. Rescale Rx and use the Q^2 calculation in theorem 2.2. Absolute convergence in the initial half-plane justifies every sum. No coprimality condition is used in this decomposition. $\square$

Define the compensated grid transform by

$$\mathcal{J}_{d,e}^{p,q}(w, B) = \int_0^\infty x^{w-1} e^{-Bx} h_d(qx) h_e(px) dx. \quad (8.4)$$

Theorem 8.2 (Rational-scale closure at every depth). *Under the same L^2 conditions on s, t, a, b as in theorem 6.3,*

$$\begin{aligned} & \frac{1}{2\pi} \int_{\mathbb{R}} \mathcal{F}_d(s, a + ipy) \mathcal{F}_e(t, b - iqy) dy \\ &= q^{s-1} p^{t-1} \mathcal{J}_{d,e}^{p,q}(s+t-1, qa + pb). \end{aligned} \quad (8.5)$$

The finite expression for the right-hand transform is

$$\begin{aligned} \mathcal{J}_{d,e}^{p,q}(w, B) &= \Gamma(w) D_{p,q}(w, B) \\ &\quad - \sum_{k \in E_e} \alpha_{e,k} p^k \Gamma(w+k) q^{-w-k} \zeta(w+k, B/q) \\ &\quad - \sum_{j \in E_d} \alpha_{d,j} q^j \Gamma(w+j) p^{-w-j} \zeta(w+j, B/p) \\ &\quad + \sum_{j \in E_d} \sum_{k \in E_e} \alpha_{d,j} \alpha_{e,k} q^j p^k \Gamma(w+j+k) B^{-w-j-k}. \end{aligned} \quad (8.6)$$

It is holomorphic for $\operatorname{Re} w > -\nu_d - \nu_e$, $\operatorname{Re} B > 0$ after all removable cancellations. The identities are stable under arbitrary spectral and position derivatives inside the stated domains.

Proof. The scaled formula (2.2) pairs $x^{s-1} e^{-ax} h_d(x)$ with $x^{t-1} e^{-bx} h_e(x)$. Their values at qx and px contribute $q^{s-1} p^{t-1} x^{s+t-2} e^{-(qa+pb)x} h_d(qx) h_e(px)$, which proves (8.5). For $\operatorname{Re} w > 2$, expand $(Q(qx) - P_d(qx))(Q(px) - P_e(px))$. The first term is theorem 8.1; a monomial from $P_e(px)$ contributes $p^k x^k$ against $Q(qx)$ and hence the second line of (8.6). The other two lines follow in the same way. The combined kernel vanishes to order $\nu_d + \nu_e$ at zero. Its holomorphy proves all pole cancellations and the derivative assertion exactly as in theorem 6.3. $\square$

If the actual slopes are cp, cq with $c > 0$, one multiplies the result by $1/c$ after changing y to cy . Thus the theorem covers every positive rational slope ratio. At $p = q = c$ it is consistent with the ordinary $1/c$ scaling law. For an irrational ratio the Laplace identity survives, but the finite residue grid used in theorem 8.1 is unavailable; no non-reducibility claim is implied.

8.1 An explicit all-order rational-scale Stieltjes formula

For depth zero write $\mathcal{J}_{0,0}^{p,q}(w, B) = \Gamma(w) K_{p,q}(w, B)$. Then (8.6) simplifies to

$$\begin{aligned} K_{p,q}(w, B) &= D_{p,q}(w, B) - \frac{p^{-1} q^{1-w} \zeta(w-1, B/q) + q^{-1} p^{1-w} \zeta(w-1, B/p)}{w-1} \\ &\quad + \frac{B^{2-w}}{pq(w-1)(w-2)}. \end{aligned} \quad (8.7)$$

For integers $m, n \geq 0$ define

$$I_{m,n}^{p,q}(a, b) = \frac{1}{2\pi} \int_{\mathbb{R}} G_m(a + ipy) G_n(b - iqy) dy.$$

Then the theorem gives

$$I_{m,n}^{p,q}(a, b) = (-1)^{m+n} m! n! [u^m v^n] q^u p^v C(u, v) K_{p,q}(1 + u + v, B). \quad (8.8)$$

The exponential slope factors are essential even for first-order jets.

For an entirely pole-free coefficient recipe put

$$z_j(c) = \frac{\zeta^{(j)}(0, c)}{j!}, \quad g_j(c) = \frac{(-1)^j \gamma_j(c)}{j!}, \quad e_j(\lambda) = \frac{(-\log \lambda)^j}{j!}. \quad (8.9)$$

Let $K_{p,q}(1 + \epsilon, B) = \sum_{k \geq 0} \kappa_k^{p,q}(B) \epsilon^k$.

Corollary 8.3 (Finite coefficient formula). *For each integer $k \geq 0$,*

$$\begin{aligned} \kappa_k^{p,q}(B) = & \frac{1}{R} \sum_{i=0}^{p-1} \sum_{j=0}^{q-1} \left\{ \sum_{\ell=0}^k e_{\ell}(R) [z_{k-\ell}(c_{ij}) + (1 - c_{ij})g_{k-\ell}(c_{ij})] + (1 - c_{ij})e_{k+1}(R) \right\} \\ & - \frac{1}{p} \sum_{\ell=0}^{k+1} e_{\ell}(q)z_{k+1-\ell}(B/q) - \frac{1}{q} \sum_{\ell=0}^{k+1} e_{\ell}(p)z_{k+1-\ell}(B/p) \\ & - \frac{B}{R} \sum_{\ell=0}^{k+1} e_{\ell}(B). \end{aligned} \quad (8.10)$$

Together with (8.8), this evaluates every rational-scale Stieltjes pairing by finitely many one-variable Hurwitz jets, Stieltjes constants, elementary logarithms, and universal zeta constants from C .

Proof. Expand (8.7) at $w = 1 + \epsilon$. The first double sum contains a simple zeta pole multiplied by $R^{-1}e^{-\epsilon \log R}$; its contribution to the coefficient of ϵ^k is exactly $(1 - c_{ij})e_{k+1}(R)/R$ in addition to the regular convolution of Taylor coefficients. The two crossed terms contain $e^{-\epsilon \log q} \zeta(\epsilon, B/q)/\epsilon$ and its p analogue. The last term is $-(B/R)e^{-\epsilon \log B}/\{\epsilon(1 - \epsilon)\}$. Their coefficients are the last two lines of (8.10).

To check the cancellation preceding extraction, observe that

$$\sum_{i,j} (1 - c_{ij}) = \frac{p+q}{2} - B.$$

The total residue of (8.7) is therefore

$$\frac{(p+q)/2 - B}{pq} - \frac{1}{p} \left(\frac{1}{2} - \frac{B}{q} \right) - \frac{1}{q} \left(\frac{1}{2} - \frac{B}{p} \right) - \frac{B}{pq} = 0.$$

This also checks the powers of p, q in the coefficient formula. $\square$

For $p = q = 1$, (8.10) reduces exactly to (3.6). At general p, q , Stieltjes jets at the grid points enter only up to order $m + n$, while zeta order derivatives at B/p and B/q enter only up to order $m + n + 1$.

8.2 The two-scale quarter-Gamma identity

The case $k = 0$ of (8.10) is

$$\begin{aligned} K_{p,q}(1, B) = & \frac{1}{R} \sum_{i,j} \{ \zeta(0, c_{ij}) - (1 - c_{ij})\psi(c_{ij}) - (1 - c_{ij}) \log R \} \\ & - \frac{1}{p} \zeta'(0, B/q) - \frac{1}{q} \zeta'(0, B/p) + \frac{\log q}{p} \zeta(0, B/q) + \frac{\log p}{q} \zeta(0, B/p) \\ & + \frac{B}{R} (\log B - 1). \end{aligned} \quad (8.11)$$

For $p = 1, q = 2, a = b = 1/2$, the grid is $3/4, 5/4$ and $B = 3/2$. Using $\psi(5/4) - \psi(3/4) = 4 - \pi$, the first line reduces to $-\pi/8$. Thus

$$K_{1,2}(1, 3/2) = -\frac{\pi}{8} - \zeta'(0, 3/4) - \frac{1}{2} \zeta'(0, 3/2) - \frac{1}{4} \log 2 + \frac{3}{4} \{\log(3/2) - 1\}.$$

Insert Lerch's formula for $\zeta'(0, z)$, $\Gamma(3/2) = \sqrt{\pi}/2$, and $\Gamma(1/4)\Gamma(3/4) = \pi\sqrt{2}$. The result is (1.6), proving that identity without numerical recognition of its constants.

9 Evaluating removable integer values without numerical cancellation

The formulas for deep compensation contain many singular summands even when their sum is regular. A stable evaluation should compute their Laurent constant terms, retaining derivatives of analytic prefactors. The following rules are enough for every integer specialization of (6.13) and (8.6).

Let $H_0 = 0$ and $H_n = \sum_{j=1}^n 1/j$. For $n \geq 0$,

$$\Gamma(-n + \epsilon) = \frac{(-1)^n}{n!} \left\{ \frac{1}{\epsilon} + H_n - \gamma + O(\epsilon) \right\}, \quad (9.1)$$

$$\text{CT}_{\epsilon=0}[\Gamma(-n + \epsilon)\zeta(-n + \delta + \epsilon, C)] = \frac{(-1)^n}{n!} \{ \zeta'(-n + \delta, C) + (H_n - \gamma)\zeta(-n + \delta, C) \}, \quad (9.2)$$

provided the zeta factor is regular there. The needed unshifted grid terms have $\delta = 0, -1$, so this proviso always holds when their Gamma factor has a nonpositive integer argument. If a regular Gamma factor meets the zeta pole instead, $k \geq 1$ and $k + \delta = 1$, then

$$\text{CT}_{\epsilon=0}[\Gamma(k + \epsilon)\zeta(1 + \epsilon, C)] = \Gamma(k) \{ \psi(k) - \psi(C) \}, \quad (9.3)$$

with residue $\Gamma(k)$. The elementary terms have

$$\text{CT}_{\epsilon=0}[\Gamma(-n + \epsilon)C^{n-\epsilon}] = \frac{(-1)^n C^n}{n!} \{ H_n - \gamma - \log C \}. \quad (9.4)$$

All these formulas follow by multiplication of one-term Laurent and Taylor expansions.

The prefactor rule is indispensable. If a singular factor has residue r and constant term c , and $\theta(\epsilon) = \theta_0 + \theta_1\epsilon + O(\epsilon^2)$, then

$$\text{CT} \left[\theta(\epsilon) \left(\frac{r}{\epsilon} + c + O(\epsilon) \right) \right] = \theta_0 c + \theta_1 r. \quad (9.5)$$

For example, q^{-w} contributes the logarithmic derivative $-\log q$. Dropping the second term in (9.5) would erase the very slope logarithms required by (8.10).

9.1 Finite algorithm and its justification

To evaluate $\mathcal{J}_{d,e}^{p,q}(w_0, B)$ at an integer w_0 inside its holomorphic domain, expand each summand of (8.6) only through its residue and constant term. For grid summands, use (9.2) or (9.3), and include the factor R^{-w} by (9.5). For crossed terms, shift w_0 by the integer monomial exponent and include the corresponding q^{-w} or p^{-w} factor. For elementary terms, use (9.4). Sum the resulting residues and constant terms. theorem 8.2 proves the residue sum is exactly zero and that the constant sum is the value of the ordinary convergent integral. Thus a finite part is used solely as a computational operation on a meromorphic formula whose total singularity is removable.

The function `J_at_integer` in the accompanying code implements this procedure, returning both the finite value and the residual sum. It avoids taking tiny spectral offsets and subtracting enormous nearly equal quantities. First-order Hurwitz order derivatives suffice for these integer values; higher spectral derivatives are obtained by retaining more Laurent coefficients. This gives an implementable extension of the primitive calculus rather than leaving its resonances as formal limits.

10 Audit findings, corrections, and verification status

10.1 Scope of the source review

The inspected canonical convolution foundation in [1] works with the mean-zero periodic family having nonzero Fourier coefficients $(2\pi i k)^{-\alpha}$. Its contact and finite-part conventions belong to that geometry.

They have not been transferred silently to the real vertical line. In the present setting, the elementary term subtracted from $\zeta(s, z)$ is $z^{1-s}/(s-1)$, and the analytic mechanism is a Laplace kernel on $(0, \infty)$.

No new false theorem was established in the canonical source portions actually inspected. It would therefore be inappropriate to issue a blanket erratum against the existing manuscript. The practical correction found during this continuation concerns the complex-argument numerical backend described next. We also isolate three normalization mistakes that would invalidate an attempted extension of the earlier calculus:

1. Subtracting only the spectral pole $1/(s-1)$ does not remove spatial growth. At $s = 1$ it leaves $\gamma_0(z) = -\psi(z)$; the squared vertical norm diverges because $\psi(a+iy)$ has logarithmic growth. Adding $\log z$ produces the convergent G_0 pairing.
2. The zero $\mathcal{R}_d(-n, z) = 0$ does not imply that its spectral derivative vanishes. The pole of $\Gamma(s)$ turns that derivative into the nonzero value in (6.10). The Stirling norm illustrates the corresponding zero-times-pole cancellation in the paired expression.
3. Replacing analytic $\log \Gamma(z)$ by the principal logarithm of $\Gamma(z)$ changes its branch by discontinuous multiples of $2\pi i$. It destroys the Laplace representation and need not preserve any vertical norm in this paper.

These are explicit scope guards, not allegations that the inspected canonical chapters made these mistakes.

10.2 A reproduced mpmath 1.3.0 nonreal-argument failure

During the independent numerical replay, the installed `mpmath 1.3.0` function `stieltjes(0, z)` failed the exact calibration identity $\gamma_0(z) = -\psi(z)$ at $z = 0.8 + 0.6i$. At 60 decimal digits of working precision it returned approximately

$$0.4662545368664062120 - 0.9931920114430890458i,$$

whereas $-\psi(z)$ is approximately

$$0.4278547847547755586 - 1.0184230001203705136i.$$

The absolute discrepancy was approximately 0.04594718437366742995, not a last-digit effect. The version-tagged source [7] takes a real part inside its integrand. That real-parameter formula is not a valid analytic continuation in a nonreal argument.

The repair used here is explicit. Complex Stieltjes values are checked by the holomorphic paired Hermite formula (3.17) and by Cauchy coefficients of the entire function $u \mapsto Z_{1+u}(z)$. Positive-real arguments in (3.6) may still use the installed Stieltjes routine; an independent positive-real control was included. The code does not claim this finding is new, that later mpmath versions share it, or that any particular canonical repository result was affected. A corpus-wide search for such unsafe nonreal calls would be a separate audit task.

For the Cauchy check, with $0 < \rho < 1$ and integer $N > m$, the approximation used is

$$G_m(z) \approx \frac{(-1)^m m!}{N \rho^m} \sum_{k=0}^{N-1} Z_{1+\rho \omega_k}(z) \omega_k^{-m}, \quad \omega_k = e^{2\pi i k/N}. \quad (10.1)$$

The actual replay used $\rho = 1/8$ and $N = 96$ at 60-digit precision for orders zero through two. This is a trapezoidal discretization of Cauchy's coefficient formula, not a certified error enclosure. Its agreement with the independent Laplace and Hermite evaluations is recorded separately from the analytic proof.

10.3 What the verification establishes

The supplied exact checks contain **538 passing assertions**: 7 kernel-valuation certificates, 36 compensated spectral-zero identities, 210 equal-scale residue cancellations, 234 rational-scale residue cancellations, 49 finite-lattice polynomial identities, and 2 Stirling-polynomial reductions. They use exact rational arithmetic and symbolic Bernoulli polynomials. These finite checks would detect many sign, indexing, or scale errors, but the proofs above, not the finite range of tests, establish the all-order assertions.

The numerical replay contains **76 passing comparisons** at **60 decimal digits of working precision**, using acceptance criterion

$$\frac{|\text{left} - \text{right}|}{\max(1, |\text{left}|, |\text{right}|)} < 10^{-35}.$$

The largest recorded scaled discrepancy was approximately 2.04×10^{-53} , in a basic spectral moment with $w = 0.8$, $A = 1$. The test set includes direct vertical-line quadrature, independent Laplace moments, mixed Stieltjes jets, complex Cauchy coefficients, polygamma integrals, rational scales, negative-order compensated moments, removable integer values, primitive derivatives, Lerch confluence, root-of-unity Gamma reductions, and finite harmonic sums. Four of these comparisons form the separate complex-safe replay; its negative control reproduces the dependency limitation just discussed.

These are unvalidated floating-point comparisons, not interval proofs. For direct vertical integrals, the implementation switches to the standard asymptotic expansions of G_0 and Ω beyond $|z| = 60$ to avoid catastrophic subtraction at very large quadrature nodes. The resulting quadrature remains a diagnostic. No Lean or other proof-assistant formalization is supplied.

10.4 Reproduction and proposed integration

The ZIP contains the modular article sources, a standalone TeX source, the compiled PDF, exact and numerical scripts, machine-readable results, a version-specific negative-control record, source audit, proof-status ledger, build instructions, and checksums. The core commands are

```
python verification/exact_checks.py
python verification/numerical_checks.py core
python verification/numerical_checks.py extensions
python verification/complex_safe_replay.py
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

A suitable integration is a new chapter or research note on *vertical-line Hurwitz pairings*, placed near the existing integration and log-Gamma material, not merged into the periodic contact-term chapter as if its integrals had the same meaning. Suggested labels and dependencies are in `INTEGRATION_NOTES.md`. The incoming ZIP should be retained unchanged as a provenance artifact, consistent with the repository's incoming-report policy. No repository write was performed in preparing this package.

11 Further research questions

The proved results suggest the following concrete extensions. These are questions, not additional theorems claimed by the present paper.

Research question 1 (Irrational-scale closure and its precise obstruction). For $p/q \notin \mathbb{Q}$, classify when $\int_0^\infty x^{w-1} e^{-Bx} h_d(qx) h_e(px) dx$ admits a finite representation by terms of the form $\Gamma(w+j) \lambda^{-w} \zeta(w+j, C)$ and elementary Gamma-power terms, with parameters independent of w . The allowed coefficient field and allowed scales must be fixed before asserting a non-reduction theorem. The Laplace identity is already available; what is missing is a rigorous classification of the corresponding exponential-kernel algebra.

Research question 2 (Joint color coalescence and Bernoulli compensation). Develop a compensation hierarchy for $(1 - \xi e^{-x})^{-1}$ that remains uniform as $\xi \rightarrow 1$ and produces well-defined mixed color/spectral jets. Can its exact transition terms be written in the same finite Hurwitz alphabet as (8.10)? A formula should specify the order of color coalescence and spectral differentiation rather than rely on formal substitution into a singular divided difference.

Research question 3 (Higher products and multiple-zeta closure). For three or more factors with both vertical orientations, Plancherel produces convolution constraints rather than a single diagonal product. Determine the smallest class of multiple Hurwitz or Tornheim-type functions closed under the resulting compensated integrals, and identify integer resonances where that class collapses to ordinary Hurwitz order jets. The two-factor result here provides exact normalization tests for such a theory.

Research question 4 (Minimal arithmetic bases for rational scales). The pq -point grid is explicit but not always minimal. Use Gamma multiplication, reflection, and finite character decompositions to find minimal provable spanning sets for $I_{m,n}^{p,q}(a, b)$ at rational a, b . The quarter-Gamma formula (1.6) is the first simple instance. Any assertion of a basis must distinguish a proved spanning statement from unproved linear independence of its constants.

Research question 5 (Exact reductions of remaining integral conjectures). Determine which still-conjectural polylogarithm integrals in the broader repository can be transformed into the finite-subtraction kernel algebra of this paper. A successful reduction would need an explicit transform identity and branch/convergence proof. No such reduction of S_6 or the revised S_8 candidate is supplied here, and numerical agreement alone would not constitute one.

Research question 6 (Certified complex jets and formalization). Replace the floating-point Cauchy replay with an interval-certified complex-zeta coefficient engine, including explicit contour truncation and rounding errors. In parallel, formalize the bilinear Laplace–Plancherel lemma, compact-parameter domination, and finite kernel identity. These three components would turn the present proof ledger into a reusable formal route to arbitrary-depth and rational-scale Stieltjes identities.

Conclusion. The central mathematical outcome is finite exact closure for a genuinely different integration geometry from the existing periodic theory. Spatial compensation gives all Stieltjes indices; deeper Bernoulli compensation gives Gamma remainders and normalized primitives; commensurate vertical scales give finite arithmetic grids; and colored kernels give explicit polylogarithm order identities. The difficult bookkeeping points—joint parameter holomorphy, removable poles, slope logarithms, and complex branches—are part of the proofs, not delegated to a numerical fit.

References

- [1] V. Reshetnikov, *ProveIt*, public research repository, canonical polylogarithm manuscript and incoming-report directory. Inspected source paths and blob identifiers are recorded in SOURCE_AUDIT.md. <https://github.com/VladimirReshetnikov/ProveIt>.
- [2] NIST Digital Library of Mathematical Functions, §25.11, *Hurwitz Zeta Function*, especially the integral representation, special values, and parameter-differentiation formulas. <https://dlmf.nist.gov/25.11>.
- [3] NIST Digital Library of Mathematical Functions, §5.9, *Integral Representations*, including Binet's formula and digamma integrals. <https://dlmf.nist.gov/5.9>.
- [4] NIST Digital Library of Mathematical Functions, §25.14, *Lerch's Transcendent*. <https://dlmf.nist.gov/25.14>.
- [5] M. A. Shpot, M. P. Chaudhary, and R. B. Paris, *Integrals of products of Hurwitz zeta functions and the Casimir effect in ϕ^4 field theories*, arXiv:1603.00722 (2016). <https://arxiv.org/abs/1603.00722>.
- [6] M. A. Shpot and R. B. Paris, *Integrals of products of Hurwitz zeta functions via Feynman parametrization and two double sums of Riemann zeta functions*, arXiv:1609.05658 (2016). <https://arxiv.org/abs/1609.05658>.
- [7] The mpmath developers, *mpmath*, version 1.3.0, `mpmath/functions/zeta.py`, function `stieltjes`. Version-specific source inspected for the nonreal-argument diagnostic. <https://github.com/mpmath/mpmath/blob/1.3.0/mpmath/functions/zeta.py>.